

Cybersecurity Incident Classification - Machine Learning Analysis

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1 Introduction

Many organizations are now facing more security incidents every day, which means that Security Operations Centers (SOCs) are receiving large number of alerts, but not all the alerts are actual real threats. Thus, it is mandatory to accurately identify which alerts correspond to real threats, or true positive (TP) incidents, as this can minimize unnecessary investigations and help analysts prioritize responses more effectively. Understanding these factors that influence the correct classification of security incidents is both practical and academically interesting. In this research paper, we aim to answer the following question:

What factors most significantly influence the accurate classification of security incidents in large-scale cybersecurity alert datasets?

To answer this main question, we define three specific subsidiary questions:

1. Out of the 6 feature groups, Organization, Detection, Entity, Account, Network, and Location, which one has the greatest impact on predicting TP incidents?
2. Which MITRE ATT&CK techniques are most frequently used together in TP incidents, and which underlying technique patterns can be identified within the dataset?
3. How do time-based features impact the classification performance for incident detection?

For this research, we used the Microsoft Security Incident Prediction dataset (Freitas et al., 2024). The dataset originates from production SOC telemetry in Microsoft Defender XDR and Copilot for Security, covering two weeks of large-scale enterprise activity; it is the largest public dataset of real-world cybersecurity incidents (Freitas et al.,

2024). For this reason, this dataset was chosen for our research because it offers a wide range of data from Microsoft, which is a reputable and verified source. Its data is authentic and not artificially synthesized by any AI tools. Using this dataset, we hope to gain a better understanding of the factors that influence incident classification and demonstrate how machine learning can help SOCs prioritize alerts more effectively.

2 Background

2.1 Dataset Terminologies

Freitas et al. (2024) established three hierarchies of data within the dataset: evidence is metadata associated with an entry in the dataset, evidence supports an alert (Freitas et al., 2024); alerts typically involve multiple instances of evidence, and alerts suggest possible security incidents. Incidents are comprised of one or more alerts, and they detail a possible threat (Freitas et al., 2024).

Incidents are then graded: a TP represents “a malicious action” that has been correctly classified, a benign positive (BP) is a “real, but not malicious [activity], such as a penetration test or known activity generated by an approved application” (Corporation, a) and a false positive (FP) is an incorrect classification of “benign content as malicious” (of Standards and Technology).

2.2 MITRE ATT&CK Framework

The MITRE ATT&CK framework improves the process of interpreting security data by providing a classification system of tactics and techniques, contextualizing the data into meaningful attack patterns for analysts and researchers (Corporation, c).

The columns of the matrix represent the high-level aims of the attacker (Fig:1). The techniques, the processes threat-actors may take to achieve the respective tactics, are the rows in each column.



Figure 1: MITRE ATT&CK Framework (Corporation, b)

3 Literature Review

3.1 Feature Group Importance Literature

Leevy et al. (2021) used the large CSE-CIC-IDS2018 intrusion detection dataset to study the impact of feature selection across several classifiers, those of which included a Decision Tree (DT) and a Naive Bayes (NB) model (Leevy et al., 2021). On the other hand, Vu et al. (2019) addressed another large-scale alert prioritization dataset using a hybrid ensemble model that combined several state-of-the-art models to predict which cybersecurity alerts required immediate attention (Vu et al., 2019). Their approach also combined distinct features into specific subsets based on the network and system logs they retrieved. Consequently, they were able to obtain an AUC score of 0.93, which demonstrated that varied, well-engineered feature groups can help to improve the model’s robustness and detection ability.

3.2 MITRE Technique Co-Occurrence Literature

Rahman and Williams (2022) explore patterns of adversarial behavior through Frequent Itemset Mining, identifying which combinations of techniques appear most frequently per adversary. Abo-alian et al. (2025) evaluate the priority of each technique based on different factors, such as relevance to real-world threats, and how well current security systems can detect it. Additionally a case study is performed, demonstrating that the highest-rank techniques directly correspond to threat group APT3’s known attack patterns, justifying that the modeling of techniques, if done with context, is relevant to real-world adversaries (Abo-alian et al., 2025). ATT&CK mappings, detector and entity context in the dataset allow frequent-itemset mining of co-occurring techniques (Freitas et al., 2024).

3.3 Time-Series Modeling Literature

As highlighted in the review by Landauer et al. (2024), time-series modeling can be utilized to predict future occurrences of cybersecurity attacks and vulnerabilities, hence enhancing the overall robustness of the system. Moreover, incorporating time-based features, such as the day of the week, month of the year, and holidays, can help capture seasonality and temporal patterns in the data, which significantly improves the predictive accuracy of time-series models. Within the two-week horizon, adding time-based features supports controlled tests of short-term temporal patterns without changing other pre-processing (Freitas et al., 2024).

4 Methods

4.1 Dataset Overview

The Microsoft Security Incident Prediction dataset (Freitas et al., 2024) contains over 13 million pieces of telemetry evidence across 33 distinct entity types, covering approximately 1.6 million alerts and 1 million annotated security incidents. Each incident includes ground truth labels that indicate if it was a TP, BP, or FP incident. The dataset contains incidents across 6,100 organizations and includes 441 MITRE ATT&CK techniques.

4.2 Feature Engineering and Selection

4.2.1 Feature Group Importance Analyses

The Microsoft Security Incident dataset contains 44 features in total. To make the analysis more interpretable and manageable, we grouped related features into six main categories:

- **Organization: OrgID**
The organization associated with the incident may indicate certain risk patterns or alert volumes.
- **Detection: DetectorId, AlertTitle, Category**
They encode what triggered the alert and how it was categorized.
- **Entity: EntityType, EvidenceRole, Roles, OS-Family, OSVersion**
This group encompasses the type of entity involved in an alert, which could either be an user, a machine, an IP, etc. OS information is included because some threats target specific platforms.
- **Account: AccountSid, AccountUpn, AccountObjectId, AccountName**

The account associated with the alert can provide context on the user’s activity.

- Network: IPAddress, Url, ThreatFamily
These features capture the external and internal network activity and potential malware families.
- Location: CountryCode, State, City
The location of an alert or evidence that can provide hints about regional attacks.

After careful analysis, the other 25 features were omitted for several reasons. The MitreTechniques and Timestamp features were already being investigated in the second and third research questions. Other features such as Id, IncidentId, AlertId, NetworkMessageId, EmailClusterId, and Sha256 are all unique identifiers. The features ActionGrouped, ActionGranular, ResourceType, and AntispamDirection were removed because most of these contains null values. RegistryValueData, RegistryKey, ApplicationId, ApplicationName, OAuthApplicationId, DeviceId, DeviceName, ResourceIdName, FileName, FolderPath, RegistryValueName contains mostly constant values, which have little to no variance and cannot help the model distinguish between TP and FPs. Finally, SuspicionLevel and LastVerdict were eliminated because they contain information about the final ground truth labels and would cause data leakage issues.

4.2.2 MITRE Technique Analyses

Following suit from Abo-alian, Youssef, and Badr’s 2025 study, to ensure industry relevance we feature engineered their open-source catalogue (all_techniques.csv) of techniques and their respective further documentation (OxmahmoudJo0, 2025; Abo-alian et al., 2025).

The following columns were chosen to derive a risk score dictionary to be used in our KM model: Technique_ID, Name, Description, x_mitre_impact_type, x_mitre_detection, x_mitre_defense_bypassed, x_mitre_permissions_required, and x_mitre_data_sources. The risk score was determined by a heuristic to indicate the priority of the technique. The heuristic is as follows:

$$\begin{aligned} \text{Risk Score} = & 0.6 \times \text{severity} \\ & + 0.4 \times \text{undetectability} \\ & + 0.2 \times \text{unmitigatability} \end{aligned} \quad (1)$$

The background and literature review for this research question informed the variables within

the heuristic. Firstly, undetectability was determined by how many data sources could detect the technique, as detect is a “critical component of modern cybersecurity strategy” (Abo-alian et al., 2025), therefore critical aspect that must be taken into account when judging technique risk. Our variable, “undetectability”, was the inverse of the detection_score, which was the sum of two normalised numeric features data source count and detect keyword count. Data source count was an amount of data sources listed in the catalogue column x_mitre_data_sources, whilst detect keyword count checked the descriptive catalogue column x_mitre_detection for keywords against a glossary of cybersecurity terms (Australian Cyber Security Centre, 2025). ChatGPT was used to analyze and extract terms from the glossary that were detection related. Due to the variance of keywords and the standardisation of the data source column within the MITRE ATT&CK framework, the data source count was weighted to be the dominant feature (70%), whereas the keyword account was weighted 30%.

Severity was determined by the impact penalty and the permission score. Impact penalty was based on the catalogue column x_mitre_impact_type, which listed the corresponding impact type(s) related to a technique. The CIA triad, Confidentiality, Integrity, and Availability (Fortinet, 2025) were the possible types listed, and the particular technique was penalised based on the amount of the triad present. This was because the impact of each impact type is extremely dependent on the context of the product or service in question, so to weight these differently would introduce bias. Entries with missing x_mitre_impact_type values were treated as having no explicit CIA impact. This avoids inflating severity for techniques that have no formally classified impact.

The permission score was based on the x_mitre_permissions_required column, with user, administrator, system and root as possible entries. The Analytic Hierarchy Process (AHP) was followed from (Belim et al., 2020) to weight the permissions, and an exponential progression was used for simplicity (user was weighted 1, administrator 2, system 4 and root 8). If a technique fits a CIA triad impact, and requires high privilege, it is particularly severe (Abo-alian et al., 2025), so severity is the product of normalised permission scores and impact penalties.

Unmitigatability was the inverse of mitigation

potential, a value added in the heuristic to account for how easy a technique may be to thwart or mitigate. This value was simply calculated from the `x_mitre_defense_bypassed` column, and then inverted, as the more defenses it has bypassed means it has lower mitigation. This was then normalised and inverted to produce the unmitigatability variable.

The AHP method used in permission scoring was also used for the heuristic weightings, along with exponential weighting difference, to emphasise the severity of a technique, yet still take into account the roles detection and mitigation play in prioritising a technique. Lastly, the normalised risk scores were used in the implementation of the KMeans model, to find the sum of the risk score for combinations of techniques in TPs, ultimately determining the prioritisation of the found technique patterns.

4.2.3 Time-Based Features Analyses

Initially, to utilise the dataset, the features are divided into nominal and cardinal types to ensure they are applied to the model correctly.

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

To compare time-based features in the experiment, the timestamps, representing when the incidents were logged, are separated from the dataset and converted into cyclical features (hours, days, weekdays, and months) using both sine and cosine terms. This method preserves temporal continuity and allows the classifiers to better capture temporal patterns (Vaswani et al., 2017). Additionally, the time features are included only when specified.

To prepare the data for training, categorical features are normalised to a 0–1 range using the transform method in Python, while numerical features are converted into a CSR (Compressed Sparse Row) matrix for efficient storage and computation. The numerical and categorical features are then combined using the `hstack` function, and the categorical target labels are encoded into numeric form with `LabelEncoder`, allowing the model to process them properly.

Stratified 5-fold cross-validation was applied on the training split, and hyperparameters were selected by mean accuracy using an identical protocol across models. All training and evaluation

for question three was executed on a Windows~11 system.

4.3 Models Selected

4.3.1 Feature Group Importance Analyses

Two models were selected to analyze the predictive capability of the chosen feature groups, a DT and a NB classifier. The DT model was chosen as the primary model because it can naturally handle both categorical and numerical variables without requiring extensive preprocessing. It then allows us to model complex and non-linear relationships between the predictors and the target IncidentGrade variable. To prevent overfitting, the model parameters were limited using a maximum depth of 10 and a minimum of 10 samples per split. Once the feature importance scores were calculated, they were aggregated to evaluate the relative contribution of each feature group.

Secondly, a Multinomial NB model was used as an affirmative as it was more suitable for categorical and one-hot encoded features. More importantly, the model is capable of computing the mutual information, which were then used to analyze the approximate influence of each feature group on the prediction outcomes. This provided an interpretable way to see how each group contributes to the TP incidents.

More complex models such as the Random Forest model were considered, but were not used in this question even though they are capable of achieving higher predictive accuracy. But these models sacrifice interpretability, which is more of the focus of this question.

4.3.2 MITRE ATTACK Technique Analyses

The first model chosen for the second research question was K-Means, as it is a pattern identification problem not a prediction problem. Additionally, it is unsupervised, as there is no classification label for combinations of MITRE techniques. The feature engineering for this model was done with the supplementary technique documentation, used from the open source repository from (OxmahmoudJo0, 2025; Abo-alian et al., 2025), which helped score the different technique combination's risk.

The best k-value was chosen by the silhouette method, which “measures how well a data point belongs in a cluster by computing the pairwise distances between all the objects in every cluster... the distance between the clusters, known as the silhou-

ette width, should be maximised” (Al-Shaer et al., 2020). Therefore, the k value that produces the largest silhouette width was chosen.

4.3.3 Time-Based Feature Impact Analyses

For the third question, Logistic Regression (LR), a linear Support Vector Machine (SVM), and a shallow Multi-Layer Perceptron (ANN) are compared to isolate temporal effects in a high-cardinality one-hot setting with additional numeric fields. This trio is chosen because K-Means is unsupervised and misaligned with label-based evaluation, Decision Trees fragment under sparse one-hot encodings and are perturbation-sensitive, and Naïve Bayes assumes conditional independence unlikely to hold. LR provides a convex linear baseline for additive, linearly separable shifts in class log-odds; a linear SVM tests margin-based separability in high-dimensional sparse regimes; a shallow MLP with ReLU captures modest non-linear interactions from cyclical time encodings.

Hyperparameters were tuned via stratified 5-fold cross-validation using mean accuracy, with grids $C \in \{0.03, 0.1, 0.3, 1, 3\}$ for SVM and LR while $\alpha \in \{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ for the MLP. Models were then refit on the full training set and evaluated once on the held-out test set. All other preprocessing and feature handling were held constant, and the procedure was executed separately for the ‘No-Time’ and ‘With-Time’ configurations.

4.4 Procedures Followed

4.4.1 Sampling and Split Statistics

A stratified sample of 100,000 incidents was drawn from the corpus and split 80% / 20% into train and test, preserving IncidentGrade (TP/BP/FP) proportions to reduce label shift across splits.

4.4.2 Data Cleaning.

For each question, targeted cleaning procedures were applied to align with the specific analysis objective. For question 1, the numerical features were converted into sparse matrices, and missing values were filled with -1. For question 2, the data was cleaned by removing rows with missing or empty MITRE technique entries. Only true positive incidents with valid MITRE technique patterns were considered, and the remaining data was filtered to include only those containing more than one technique. For question 3, timestamps were parsed to derive hour, dayofweek, and month and cyclic encodings were added, these transformed features

were appended to the numerical feature list prior to vectorization.

4.4.3 Training Protocol

Models were trained on the training split. Hyperparameters were preset, and cross-validation was used for tuning. Class imbalance was handled by stratified sampling or loss reweighting. Model selection relied on validation performance and used early stopping or a fixed epoch budget.

4.4.4 Test and Evaluation

Each model was evaluated on the test dataset. Reported metrics include Accuracy, Macro-Precision, Macro-Recall, and Macro-F1, and results were presented using related analytical plots.

4.5 Hypotheses

4.5.1 Feature Group Importance

Detection features will contribute the most to predicting TPs, followed by Entity and Account features. Network features will offer additional but smaller gains, while Organization and Location will show the weakest incremental impact. This is because the detection metadata is closest to the decision boundary learned by security products, entity and account context captures who or what was involved, and network indicators are often sparse or noisy. Organization and location mainly reflect background heterogeneity and add limited class discriminatory signal after other groups are included.

4.5.2 MITRE Technique Co-Occurrence

We hypothesize that K-Means clustering will produce similar groupings of attack techniques that correspond to adversary objectives. Al-Shaer, Spring and Christou (2020) noted that software attacks are expected to form smaller clusters, as the type of attack is reliant on fewer tactics and techniques, whilst Advanced Persistent Threat (APT) attacks would involve multiple tactics. Abo-alian, Youssef(Abo-alian et al. (2025)) and Badr (2025) evaluated their model results against an actual adversary, APT3, also known as Gothic Panda, and was able to resolve their known tactics with a pattern found through their analysis. Therefore, we hypothesize that different styles of attacks (software, APT, etc.) will be more prevalent in certain pattern sizes, and that combinations of techniques will be grouped into tactics that are commonly used together. Additionally, with the introduction of sub-techniques into the MITRE ATT&CK framework,

we hypothesize that techniques will often be paired with their sub-techniques, as they are variants.

4.5.3 Impact of Time-Based Features

Adding time-based features is expected to affect models unevenly. In high-dimensional one-hot spaces, linear SVM may lose macro-F1 and recall when temporal signals are weak, shift across splits, or interact with categorical indicators in ways that require non-linear composition. Under such conditions, margin models can absorb variance without added separability unless regularization is carefully tuned. LR can benefit when regularization accommodates the expanded design, but may also suffer if time-based features act mainly as noise. By contrast, a MLP trained on a densified and scaled design matrix can form modest non-linear interactions between time-based features and entity attributes while early stopping limits overfitting. So the expected ordering with time-based features may be $LR \approx MLP \geq SVM$, and accuracy may move less than macro-F1 and recall.

5 Evaluation

We decided to incorporate four classification evaluation metrics: accuracy, precision, recall, and F1-score to assess the model’s performance. These are all macro-averaged, as the dataset contains imbalanced classes. Specific incident grades like TPs might occur more often than others. This ensures that each class contributes equally to the overall score and provides a fair evaluation of model performance across all categories. Specifically, we focus on accuracy, which measures the overall proportion of true positive incidents and is the main focus of our research question.

5.1 K-Means Evaluation

The K-Means model performance was evaluated by cross-referencing the riskiest patterns discovered with results from the aforementioned studies by [Abo-alian et al. \(2025\)](#) and [Al-Shaer et al. \(2020\)](#). This evaluation method follows the field’s case-study standard and primarily seeks to validate the model’s findings against real adversary attack patterns.

The technique patterns discovered in our results for the top riskiest patterns align well with the results from [Abo-alian et al. \(2025\)](#) for individual techniques, notably the presence of T1105, T1204.002, verifying that our model’s clustering

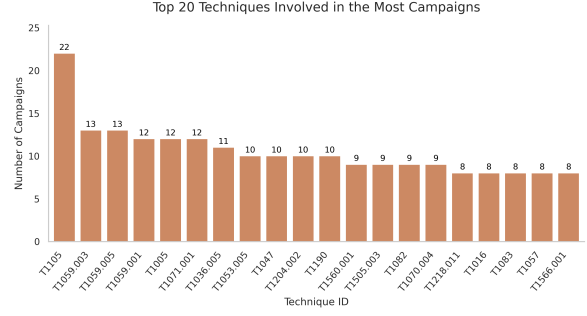


Figure 2: ([Abo-alian et al. \(2025\)](#)) Top Risky Campaign Techniques

was able to produce similar results (taking into account the differences of datasets).

6 Results

6.1 Feature Group Importance Results

From the model performance metrics reported in Table 1, the DT model achieved a higher overall performance in terms of all four evaluation metrics, accuracy, precision, recall, and F1-score compared with the NB model.

Model	Accuracy	Precision	Recall	F1-Score
DT	0.7718	0.7939	0.7337	0.7496
NB	0.4011	0.3545	0.3628	0.3471

Table 1: Performance comparison between DT and NB classifiers.

Based on the DT feature importance analysis as shown in Figure 4, the Organization group contributed the most to the prediction of TP incidents, accounting for 55.3% of the total importance. This is followed by the Detection group, contributing to 37.2%. The remaining feature groups had comparatively lower influence, with Entity (5.6%), Account (1.2%), Network (0.6%) and Location (0.03%) showing minimal impact on classification outcomes.

Surprisingly, for the NB model, the feature influence estimated using mutual information between each feature group and the target variable indicated that the Detection group contributed the highest information gain (32.9%), followed by Account (28.8%) and Organization (22.0%) as shown in Figure 3. The network and entity groups contributed slightly less at 7.3% and 4.3% respectively, with Location having the smallest influence of (4.6%) as per the DT model.

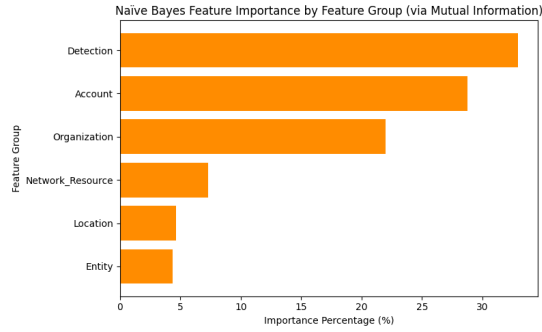


Figure 3: Comparison of approximate feature influence using NB via mutual information.

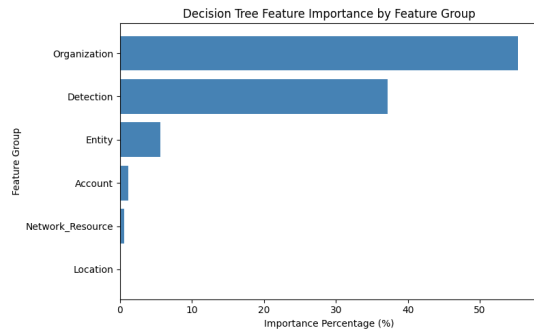


Figure 4: Comparison of feature importance using DT.

6.2 MITRE Technique Patterns Results

The most dominant tactics per cluster were: Valid Accounts for cluster 0, Brute Force for cluster 1, System Service Discovery for cluster 2, Process Injection for cluster 3, Valid Accounts (again, with variants) for cluster 4, Ingress Tool Transfer for cluster 5, and OS Credential Dumping for cluster 6 (Fig:5).

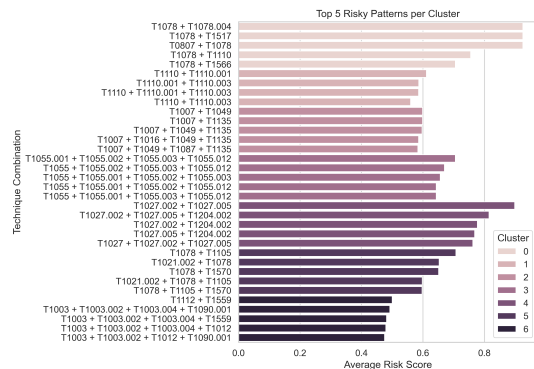


Figure 5: Top Risky Patterns per Cluster

The most severe cluster was deemed cluster 4, corresponding to the techniques T1027.002 + T1027.005 + T1204.002, Obfuscated Files or

Information: Software Packing, Obfuscated Files or Information: Indicator Removal from Tools, and User Execution: Malicious File (Fig:6).

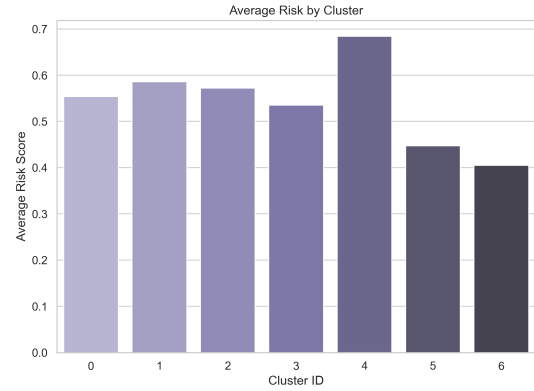


Figure 6: Risk by Cluster

The pattern size of technique combinations displayed little to no differences in the risk score (Fig:7).

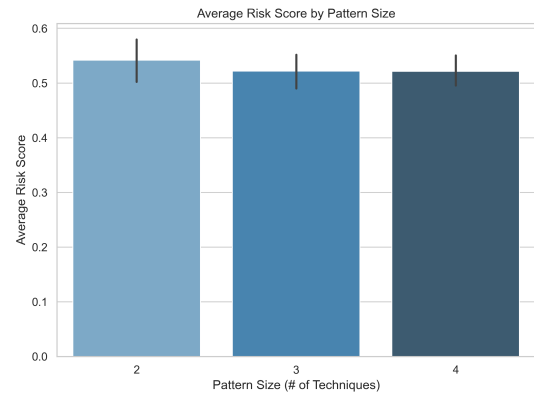


Figure 7: Risk by Pattern Size

The top riskiest patterns were the use of the Valid Accounts techniques and sub-techniques, Valid Accounts paired with Command-Line Interface, Valid Accounts with Access Notifications, and Obfuscated Files or Information paired with relevant sub-techniques (Fig:8).

6.3 Time-Based Features Results

Model	No Time	With Time
SVM	$C = 0.1$	$C = 1.0$
LogReg	$C = 1.0$	$C = 0.3$
ANN	$\alpha = 10^{-3}$	$\alpha = 10^{-5}$

Table 2: Best hyperparameters (cross-validated).

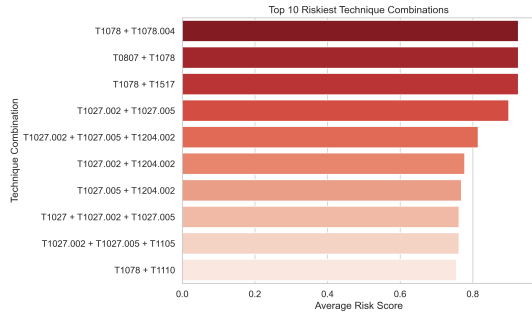


Figure 8: Top 10 Riskiest Patterns

Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1
SVM	0.5480→0.4810	0.5746→0.3197	0.4729→0.4086	0.4111→0.3587
LogReg	0.4945→0.5595	0.5189→0.5805	0.4395→0.5029	0.4290→0.4797
ANN	0.6747→0.6933	0.6905→0.6944	0.6108→0.6392	0.6148→0.6457

Table 3: Performance (No Time → With Time)

Model	Δ Accuracy	Δ Precision	Δ Recall	Δ F1
SVM	-0.0670	-0.2549	-0.0643	-0.0524
LogReg	+0.0650	+0.0616	+0.0634	+0.0507
ANN	+0.0186	+0.0039	+0.0284	+0.0309

Table 4: Performance deltas (Δ = With Time - No Time)

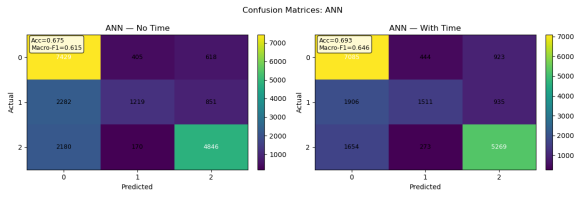


Figure 9: Confusion Matrices: ANN

Label mapping: 0 = BP, 1 = FP, 2 = TP. All subsequent Confusion Matrices figures use the same label mapping.

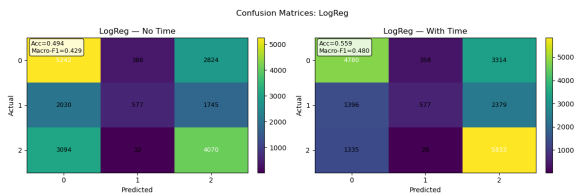


Figure 10: Confusion Matrices: LR

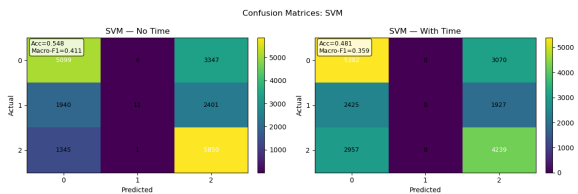


Figure 11: Confusion Matrices: SVM

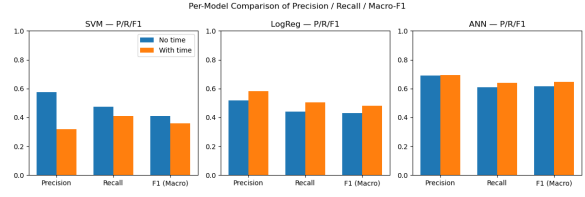


Figure 12: Comparison of Precision / Recall / Macro-F1

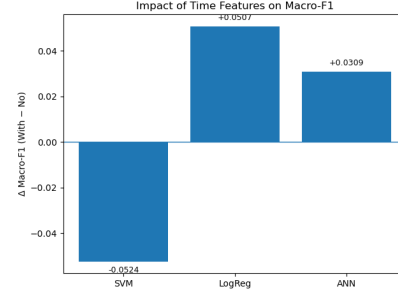


Figure 13: Impact of time features on Macro-F1.

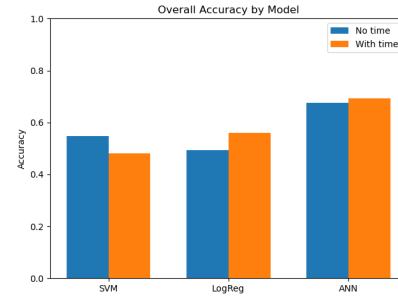


Figure 14: Overall accuracy by model

7 Discussion

7.1 Feature Group Importance Discussion

The results in (Table 1) showed that the DT model performed much better than the NB model in predicting the incident grades. The DT model obtained a substantially higher accuracy of 77% compared to the NB model, which was only 40%, in terms of precision and recall as well. This difference can explain the nature of the dataset being dependent, which means that models like DTs can help capture non-linear and conditional dependencies, ultimately being more suitable in this case. NB assumes that the features are independent and hence failed to represent the complex dependencies that are present in this cybersecurity dataset.

The feature importance analysis revealed that Organization and Detection groups were the strongest predictors, together contributing over 90% of the DT's total importance Figure 4. These results point towards the fact that both the source of an alert

and how it is detected have the greatest influence on determining whether an incident is a TP. Subsequently, other groups such as Entity, Account, Network, and Location have extremely small contributions. Upon detailed inspection of the actual values of these features, their lower influence is suspected to be due to a lack of variation, where most of the values are either from a single account or a single location.

The NB mutual information analysis vaguely supported these findings, also ranking Detection in first place and Organization in third place Figure 3. Despite emphasizing Account slightly more than Organization in second place, its overall performance can confirm that a simpler probabilistic model might struggle with this type of structured interdependent data.

7.2 Feature Group Importance Relevance

Our findings are consistent with those from [Vu et al. \(2019\)](#), which showed that diverse and context-rich feature groups improved the reliability of threat detection models. However, the results only partially confirm the hypothesis that features related to the organizational context and alert characteristics would have the greatest impact on predicting TP incidents. The hypothesis initially claimed that Entity might be the strongest predictor, but the comparatively lower influence of Entity, Account, and Location features from the results suggested that individual user or geographic information alone is insufficient to improve the classification’s accuracy. This shows that context and detection details matter more than just basic metadata.

7.3 MITRE Technique Patterns Discussion

The clustering results show that K-Means effectively identified distinct technique patterns within the Microsoft incident dataset, supporting the hypothesis that related MITRE ATT&CK techniques would appear together to achieve a specific adversary objective. Additionally, each cluster produced a dominant tactic that demonstrated the leading behavior of the cluster: cluster 0 and cluster 4 were dominated by Valid Accounts (T1078) and its sub-techniques, indicating credential misuse and lateral movement. Cluster 1 was dominated by Brute Force (T1110), while cluster 2 was mainly System Service Discovery (T1007), meaning that the adversaries gain details about local system services to first understand the target environment, then use the information to attack ([Corporation, b](#)). Clus-

ter 3 was dominated by Process Injection (T1055) patterns, and cluster 5 by Ingress Tool Transfer (T1105), meaning adversaries move tools into compromised environments, then attack. Lastly, cluster 6 was defined by OS Credential Dumping (T1003), indicating credential theft.

The top risky patterns per cluster demonstrate that different tactics correspond to different risk levels. As shown in the risk by cluster figure, cluster 4 was ranked the most risky, indicating the technique pattern creates the biggest impact, and was the hardest to mitigate. As mentioned, the cluster pattern corresponds to the tactics of Obfuscated Files or Information: Software Packing, Indicator Removal from Tools, and User Execution: Malicious File. This pattern implies that obfuscation and social engineering-based execution are used in the biggest-impact attacks. This is also consistent with the tactics found in APT campaigns ([Al-Shaer et al., 2020](#)).

The pattern size, the number of techniques in a pattern, did not significantly affect the risk score. This suggests that risk is more dependent on the specific techniques, and the realm they operate in, as stated by [Al-Shaer et al. \(2020\)](#) in noting that software attacks and APT attacks differ in technique pattern size.

Lastly, the top 10 riskiest patterns find that the Valid Accounts tactic (T1078) and its sub-techniques, often paired with Command-Line Interface (T1059) and Access Notifications (T1517), made for the most risky pattern. This implies that credential abuse (Valid Accounts) is one of the highest risk tactics used in recorded Microsoft incidents.

Additionally, patterns involving Obfuscated Files or Information (T1027) and its sub-techniques are among the top-risk patterns, meaning that avoiding detection behaviors (like the sub-techniques present in our results, such as tool packing and indicator removal) also has a large impact on Microsoft.

7.4 MITRE Technique Patterns Relevance

The relevance of the resulting MITRE ATT&CK technique patterns is bolstered by the evaluation performed against the 2025 results in the study by [OxmahmoudJo0 \(2025\)](#); [Abo-alian et al. \(2025\)](#). Additionally, the importance of understanding the MITRE ATT&CK technique patterns is confirmed by [Yulianto et al. \(2025\)](#), as their 2025 study states that “the integration of the MITRE ATT&CK

framework into red-teaming exercises represents a significant advancement in the field of cybersecurity". Yulianto et al. (2025) detail the numerical effectiveness of applying MITRE ATT&CK technique pattern knowledge in the industry, noting a 18% reduction in security incidents annually, a 30% decrease in undiscovered vulnerabilities, and importantly, a 20% decrease in successful cyber-attacks. Additionally, a 40% increase in organizations adopting MITRE ATT&CK was reported by Yulianto et al. (2025), indicating that the analysis of MITRE ATT&CK technique patterns is worthwhile, and is becoming more prevalent. Ultimately, MITRE ATT&CK technique patterns identified from analysis can inform and improve a security team's effectiveness and approaches.

7.5 Time-Based Features Discussion

Model selection uses validation accuracy; the best settings with and without time for each classifier are summarized in Table 2.

Across the three classifiers, time-based features have uneven effects (Fig. 12–13; Table 3–4). With time enabled, MLP (ANN) increases both accuracy and Macro-F1, mainly through higher Macro-Recall. LR shows the largest improvement. Linear SVM declines on both metrics.

The confusion matrices make the pattern concrete. For ANN, the FP row shifts toward the FP column and the main diagonals remain strong (Fig. 9). For LR, the TP diagonal strengthens while the FP prediction column stays small (Fig. 10). For SVM, the FP prediction column collapses with extra mass leaking into TP (Fig. 11).

The underlying mechanism is consistent with model capacity and data structure. Time-based features are weak on their own and become useful when they interact with categorical attributes. ANN, trained on a densified and scaled design with early stopping, can express such interactions and translate them into recall gains for FP. Linear models operate additively in a sparse one-hot space. Without explicit cross-terms, added time features increase variance more than separability under class imbalance, which tilts margins and log-odds toward BP and TP and reduces FP detection. Metric sensitivity follows the same logic that Macro-F1 moves more than accuracy when changes concentrate on FP (Fig. 14– Fig. 13).

Limitations

We only used 100,000 samples from a dataset of over 13 million, which could limit how well the model generalizes to the full dataset. K-Means assumes clusters are evenly sized and that all features are equally important, which might oversimplify complex patterns. We also didn't fully explore feature selection, so some less informative or noisy features may have been included. Finally, the model hyperparameters weren't extensively optimized, which could affect prediction accuracy—for example, the LR model might perform worse with more iterations when time-based features are added.

8 Conclusion

In this study, we first investigated the prediction of incident grades using selected feature groups and machine learning models. Our analysis of feature importance revealed that organizational and detection-related features contributed most to model performance, justifying the selected feature groups. These results demonstrate that careful feature selection can improve classification accuracy.

The analysis of MITRE ATT&CK technique patterns using K-Means found that the technique combinations involving credential misuse and obfuscation were the highest risk for the Microsoft incident dataset. These findings demonstrate the importance of understanding and using MITRE technique patterns to inform cybersecurity defense strategies.

While including time-based features yielded the largest improvements for LR and modest gains for the ANN classifier, the linear SVM deteriorated as added time dimensions increased variance without adding separability. Performance sensitive to hyperparameter choices and to a model capacity to capture interactions between time features and categorical attributes.

9 GenAI Usage

ChatGPT was used for developing a helper function for feature engineering research question 2. This was noted in the code, prior to the function (preprocessing, mitre-feature-engineering.py). Additionally, Chat GPT was used to determine which terms from a glossary were detection relevant, as noted in preprocessing, mitre-feature-engineering.py in the codebase.

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